

Round 1: Summary of idea

Problem/ Solution

Problem Worth Solving

CycleBox builds a cardboard return system grounded in sustainable development principles. As e-commerce and delivery services continue to grow, cardboard boxes have become one of the most common forms of packaging in urban life. People receive packages frequently, and they also frequently deal with the boxes that come with them.

Although cardboard is technically recyclable, the reality is different. Many boxes are simply thrown into regular trash, or lose their recycling value after being mixed with other waste. Even when boxes enter the recycling system, most of them are broken down into pulp instead of being reused as whole boxes. As a result, there is still significant room to improve the return rate of shipping boxes.

At the same time, companies face increasing pressure to reduce carbon emissions and meet sustainability goals. However, existing recycling systems do not effectively motivate users to participate. This leads to unnecessary material waste and greater environmental impact.

The low return rate of cardboard boxes is not only caused by a lack of environmental awareness. A key issue is the absence of practical incentive mechanisms. For most users, folding, sorting, and separately handling boxes requires extra time and effort, but provides almost no direct benefit. Cardboard boxes have little material or emotional value in users' perception.

Contamination further reduces recycling success. Contamination can occur on the user side, for example when boxes are mixed with household waste. It can also occur during waste collection or sorting if materials are not handled properly. In these cases, boxes that could have been reused lose their value.

In addition, the current recycling system and the logistics system operate separately. Recycling is mainly managed by municipal or waste management agencies, while logistics is operated by e-commerce platforms and delivery companies. There is little connection between these two systems. This structural separation makes it difficult for whole boxes to return to the logistics network. Even when boxes are collected, they usually enter the pulp recycling process rather than being reused as packaging assets.

As carbon disclosure requirements and sustainability targets become more demanding, relying only on material recycling is no longer sufficient.

The challenge is how to increase box return participation without significantly increasing operational burden, and how to explore the feasibility of whole-box reuse. This challenge represents both an environmental necessity and a potential economic opportunity.

Target Customer

CycleBox serves two layers of customers. In the early stage, our primary customers are regional e-commerce warehouse operators and regional logistics partners. Compared to large national delivery companies, regional logistics and warehousing firms are more flexible in pilot testing. They have shorter decision chains and can more easily experiment with new return mechanisms within specific communities.

These companies handle large volumes of e-commerce boxes every day. The cost of box procurement, transportation loss, and sustainability performance all directly affect their long-term operations. As companies place greater emphasis on carbon reporting and environmental performance, relying solely on material recycling is no longer sufficient to demonstrate higher levels of circular efficiency. Therefore, they have practical motivation to explore solutions that increase box return rates and enable whole-box reuse.

In the early stage, CycleBox does not require companies to fundamentally change their existing systems. Instead, we build on their current delivery networks and run small-scale community pilots to validate participation rates and reuse feasibility. For example, delivery drivers can collect qualified boxes directly from individual households during regular routes. This approach lowers the barrier to collaboration and reduces operational risk.

At the pilot level, residential communities are the key application environment. Target communities typically have stable e-commerce activity and consistent delivery frequency, such as medium-density neighborhoods or multi-unit residential areas. End users in these communities are not paying customers, but they are the most critical behavioral participants in the system.

Although online shoppers have high purchasing frequency, they lack direct incentives to return cardboard boxes. This creates clear room for behavioral improvement. By introducing reward points and emotional feedback mechanisms, CycleBox aims to change users' perception of cardboard from "low-value waste" to a meaningful asset within a circular system.

From a market perspective, the volume of e-commerce parcels in the United States remains consistently high, generating hundreds of millions of cardboard boxes annually. Even within a single city, daily delivery volume is sufficient to support small-scale pilot validation.

CycleBox focuses its early market on city-level or community-level closed-loop environments.

By increasing participation rates and validating reuse cycles, we aim to build data-based evidence for scalability. Once pilot data demonstrates that incentives can significantly improve return rates while controlling contamination and loss, the model can be replicated across additional communities and regions.

Proposed Solution

CycleBox proposes a phased circular system centered on one key question: how can we increase participation in e-commerce box returns?

The core logic is to use behavioral incentives to reshape users' perception of cardboard value, while gradually validating the feasibility of whole-box circulation within existing logistics networks. At the same time, the system is designed to control contamination and operational risk as it scales.

In Phase One, CycleBox does not introduce new hardware or reconstruct the existing logistics system. Instead, we conduct low-cost pilot programs in selected residential communities. These pilots rely on the current delivery network, where delivery drivers collect qualified boxes within designated areas during normal routes. The pilot focuses on validating three key variables: whether return participation rates significantly increase, what proportion of returned boxes meet reuse standards, and the average reuse cycle of a single box under real-life conditions.

Through data collection and graded assessment, we evaluate whether whole-box return is practically feasible in everyday environments. To increase participation, CycleBox establishes a dual-layer incentive mechanism. On the material side, we integrate with e-commerce or logistics partners' reward systems, allowing users to receive points or tangible benefits when returning boxes. This gives cardboard a clear economic value.

On the emotion side, CycleBox boxes are designed with a distinct visual identity rather than remaining plain brown cartons. For example, each box may feature a printed illustration of a stray cat or dog on the exterior surface, symbolizing that the box is part of an ongoing journey instead of disposable packaging. When a user returns the box, the system provides digital feedback through the platform, such as showing that the "stray" has been successfully returned to a safe network or shelter within the CycleBox system. This is not meant to be literal, but symbolic — reinforcing the idea that the box continues circulating within a managed return loop. The purpose of this design is to create emotional attachment and a sense of participation, turning box return from a mechanical recycling action into a meaningful and repeatable behavior..

Once pilot data becomes stable, Phase Two focuses on optimizing return pathways and grading mechanisms. For example, a whitelist system can be introduced for low-contamination product categories to increase effective reuse rates.

Under appropriate conditions, we may also explore more durable or standardized box structures to reduce transportation damage and improve reuse stability.

In Phase Three, once return density and reuse data reach a stable level, CycleBox may introduce smart collection terminals and a more structured return network. This stage aims to improve operational efficiency, reduce manual sorting costs, and enable data-driven management.

For regional logistics or warehouse partners, CycleBox provides a pathway to embed circular mechanisms into their existing delivery systems. By increasing box return participation and validating whole-box reuse, companies may reduce long-term packaging procurement dependence while generating measurable circularity data to support sustainability goals. For users, CycleBox transforms environmentally responsible behavior into tangible benefits and positive experiences, lowering the barrier to participation.

Video URL

[CycleBox](#)

By Siying Zhang, Haiyang Wu, Wandong Zhang, Lingsheng Zhou.

Round 2: Business Plan

Introduction

Problem Our Company Solves

With the rapid growth of e-commerce logistics, urban households generate a large number of cardboard boxes every day. These boxes have significant potential for reuse; however, in reality, most of them never enter an effective reuse cycle. Many boxes are discarded after a single use, while others become contaminated during collection and transportation, losing their potential for reuse.

Current recycling systems primarily focus on material recovery rather than whole-box reuse. Even when cardboard enters recycling channels, it is typically compressed and processed into pulp instead of being reused as intact boxes. For ordinary users, keeping and returning boxes requires additional time and effort, yet offers little to no direct benefit. This imbalance between effort and reward significantly reduces users' willingness to participate in box recycling.

Another challenge lies in the disconnect between urban recycling systems and logistics distribution networks. Packaging is generated at the consumer end, but there are few clear pathways for it to re-enter the logistics system for reuse. Without a structured recovery channel, whole-box recycling is difficult to achieve in practice.

These factors lead to a common outcome: a large number of boxes that still retain functional value are discarded prematurely. As a result, resource utilization remains inefficient, and additional environmental pressure is created due to the repeated production and transportation of packaging materials.

The CycleBox project aims to improve the efficiency of packaging reuse. By increasing user participation and establishing a clearer recovery process for boxes, the project seeks to support a more efficient reuse cycle for e-commerce packaging.

Background of Problem

With the rapid expansion of e-commerce, home delivery has become a routine part of urban life. According to the U.S. Census Bureau, total U.S. e-commerce sales exceeded \$1 trillion in 2023, reflecting the continued growth of online retail and the logistics systems that support it. As more goods are shipped directly to households, the use of shipping packaging—particularly corrugated cardboard boxes—has increased substantially.

Corrugated boxes are the dominant form of packaging in e-commerce logistics. Data from the U.S. Environmental Protection Agency (EPA) shows that the United States generates more than 40 million tons of container and packaging waste annually. A significant portion of this waste comes from shipping materials used in e-commerce fulfillment. Although cardboard is widely considered recyclable, the efficiency of real-world circulation remains limited.

Current recycling systems primarily focus on recovering raw materials rather than enabling the reuse of intact boxes. When cardboard enters recycling streams, it is typically compressed and processed into pulp instead of being reused as a functional shipping container. At the same time, the American Forest & Paper Association reports that the recycling rate for corrugated cardboard is approximately 68 percent. While relatively high compared with many materials, this still indicates that a large share of boxes are discarded after a single use or become contaminated during disposal and collection.

Another structural issue lies in the disconnect between municipal recycling systems and logistics distribution networks. Cardboard boxes are generated at the consumer end of the supply chain, yet there is rarely a direct pathway for those boxes to re-enter logistics operations for reuse. Without a coordinated return channel, whole-box circulation rarely occurs under normal urban conditions.

The consequences affect several stakeholders. Municipal waste management systems face increasing pressure from growing volumes of packaging waste. Logistics companies must continually produce or purchase new packaging materials, which raises operational costs and resource consumption. Consumers also experience the inconvenience of managing large amounts of packaging waste at home.

As e-commerce continues to grow, improving the circulation efficiency of shipping boxes has become an increasingly important challenge. Finding practical ways to increase return participation and enable the reuse of intact boxes represents an opportunity to reduce waste, improve resource efficiency, and support more sustainable logistics systems.

Strengthens 1. Transforms packaging from a single-use material into a reusable asset, increasing resource efficiency. 2. Directly addresses the upstream bottleneck of user participation rather than downstream processing. 3. Can be integrated into existing logistics networks without requiring new transportation infrastructure. 4. Creates clear economic value by reducing the need for new box procurement. 5. Can establish localized closed-loop systems in communities before scaling. 6. Enables tracking of box circulation data, supporting future optimization. 7. Aligns both economic incentives and sustainability goals within one system.	Weaknesses 1. Cardboard has limited physical durability, constraining the number of reuse cycles. 2. Requires users to store and return boxes, increasing behavioral burden. 3. The system depends on sufficient participation density to function effectively. 4. Lack of standardized box sizes reduces matching and reuse efficiency. 5. Requires additional inspection and sorting processes, increasing operational complexity. 6. Contamination and damage can significantly reduce the usable return rate. 7. Incentive mechanisms introduce ongoing cost pressure.
Opportunities 1. Continuous growth of e-commerce generates a large and stable supply of boxes. 2. Increasing corporate focus on ESG and sustainability initiatives. 3. Logistics companies have strong incentives to reduce packaging costs. 4. High-density environments such as campuses and urban apartments enable closed-loop systems. 5. Circular economy trends are gaining policy and market support. 6. Consumers are more willing to engage in incentivized sustainable behaviors. 7. Advances in tracking and digital systems can improve operational efficiency.	Threats 1. Traditional recycling systems are already mature and cost-efficient. 2. Low cost of new cardboard boxes reduces the economic advantage of reuse. 3. User habits are difficult to change, leading to uncertain participation levels. 4. Logistics systems prioritize efficiency and may resist additional operational steps. 5. Large platforms may develop similar in-house solutions. 6. Insufficient return rates can prevent the system from forming a viable loop. 7. Quality issues such as damage and contamination may disrupt system reliability.

Sources

U.S. Census Bureau. Quarterly Retail E-Commerce Sales.

<https://www.census.gov/retail/ecommerce.html>

U.S. Environmental Protection Agency (EPA). Facts and Figures about Materials, Waste and Recycling.

<https://www.epa.gov/facts-and-figures-about-materials-waste-and-recycling>

American Forest & Paper Association. Paper and Cardboard Recycling Rates.

<https://www.afandpa.org/priorities/recycling>

Relevant Macro and Market Trends

With the continued expansion of e-commerce, parcel delivery has become an essential part of daily urban life. An increasing share of retail purchases now reaches consumers through home delivery, which has significantly increased the volume of packaging used in logistics operations. Corrugated cardboard boxes remain the most widely used form of shipping packaging in e-commerce fulfillment. As online retail continues to grow, the number of shipping boxes entering households also continues to rise, bringing greater attention to the efficiency of packaging circulation.

At the same time, sustainability and environmental responsibility have become increasingly important priorities for governments, businesses, and consumers. Packaging waste is receiving growing public and regulatory attention as cities seek to reduce landfill pressure and improve resource efficiency. Many companies are also facing stronger expectations to reduce environmental impact across their supply chains, particularly in areas such as packaging materials and logistics operations.

In response to these pressures, the concept of a circular economy has gained traction across multiple industries. Businesses are exploring ways to extend the useful life of materials and reduce dependence on single-use resources. In logistics and packaging, improving the reuse and circulation of existing materials has emerged as a practical pathway for reducing waste while maintaining operational efficiency.

Consumer attitudes are also gradually shifting. Many consumers are more willing to participate in environmentally responsible behaviors when those actions are simple and provide visible value. Recycling programs, packaging return initiatives, and sustainability incentives have become increasingly familiar in everyday life. These behavioral shifts create opportunities for models that combine user participation with improved logistics circulation.

Customer and Market Analysis

Target Market

CycleBox operates within the rapidly expanding e-commerce logistics ecosystem, where large volumes of shipping boxes are generated every day. The target market includes both logistics operators that manage parcel delivery and urban consumers who regularly receive e-commerce packages. Together, these groups form the operational environment where cardboard box circulation can be improved.

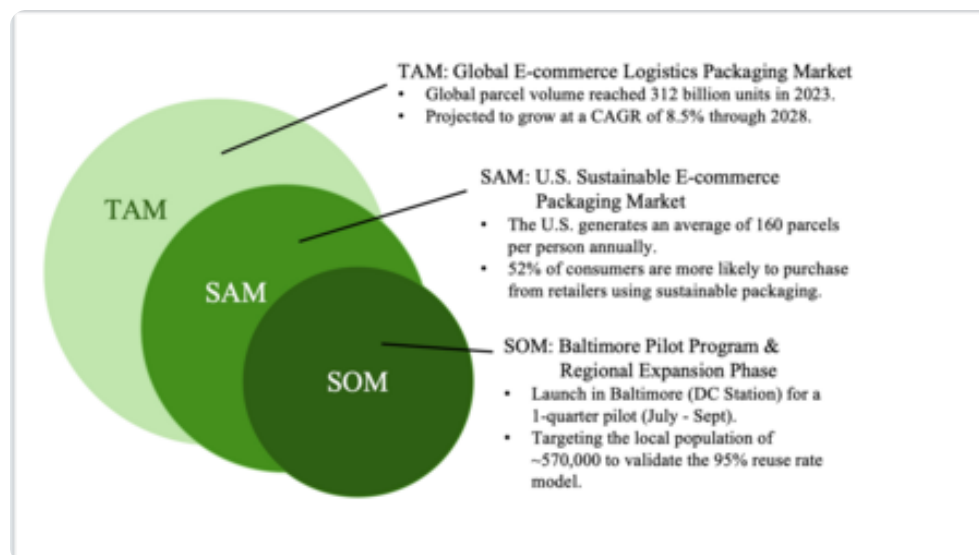
From a commercial perspective, the primary customers are logistics companies, regional delivery operators, and e-commerce fulfillment networks that handle large parcel volumes. These organizations continuously purchase corrugated packaging materials to support shipping operations. As parcel volumes grow, packaging procurement becomes a significant and recurring operational cost. Improving the reuse rate of shipping boxes offers logistics companies an opportunity to reduce packaging expenses while supporting corporate sustainability and supply chain efficiency goals.

The scale of the parcel delivery market illustrates the size of this opportunity. According to the Pitney Bowes Parcel Shipping Index, the United States shipped approximately 22.4 billion parcels in 2024, generating more than \$203 billion in parcel shipping revenue. Parcel volume is projected to continue growing and may approach 30 billion shipments annually by 2030. This massive logistics flow creates a substantial volume of cardboard packaging entering residential areas every year.

From the user side, the secondary segment includes urban consumers who regularly receive online orders. The average U.S. household now receives over 160 packages per year, which means large numbers of cardboard boxes accumulate at the consumer end of the supply chain. However, most consumers currently lack a convenient or rewarding mechanism to return shipping boxes for reuse, leading to low participation in packaging return behavior.

At the global level, the corrugated packaging market exceeds \$190 billion, reflecting the overall scale of shipping box production and usage. Within this broader market, CycleBox focuses specifically on packaging used in e-commerce logistics, where parcel volumes are high and box turnover is frequent. In the early stage, however, CycleBox is not targeting the entire market. Instead, the initial focus is on a much smaller and more defined segment, starting with pilot testing in high-density residential areas such as the Johns Hopkins University community in Baltimore. This environment provides a concentrated user base and frequent delivery activity, making it suitable for testing box return behavior and operational feasibility. Following successful validation, CycleBox plans to expand to similar urban communities and gradually partner with regional e-commerce and logistics platforms, such as Weee!, to scale the system.

By focusing on regions with high e-commerce penetration and established delivery routes, CycleBox aims to validate packaging return behavior and demonstrate the feasibility of whole-box circulation within existing logistics networks. As parcel volumes continue to increase, improving the reuse rate of shipping boxes represents a scalable opportunity within the broader packaging and logistics industry.



Pitney Bowes. Parcel Shipping Index.

<https://www.pitneybowes.com/us/shipping-index.html>

Pitney Bowes Investor Relations – U.S. Parcel Shipping Index

<https://www.investorrelations.pitneybowes.com>

American Forest & Paper Association – Corrugated Packaging Industry Data

<https://www.afandpa.org>

Global Market Insights – Corrugated Boxes Market Size

<https://www.gminsights.com/industry-analysis/corrugated-boxes-market>

Size of Initial Target Segment

The initial pilot of CycleBox will focus on a concentrated urban area around Johns Hopkins University in Baltimore. This area was selected due to its high population density, strong e-commerce usage, and manageable geographic scope for operational testing.

The target population is estimated at approximately 8,000 to 12,000 potential users, including students and nearby residents. University environments typically have higher online ordering frequency, making them suitable for testing packaging return behavior.

Assuming a conservative participation rate of 20% to 30%, the pilot is expected to engage around 1,600 to 3,600 active users during the three-month period.

In terms of order volume, if each active user places an average of 2 to 4 orders per month, the total number of deliveries during the pilot is estimated at 9,600 to 43,200 orders over three months.

Since each order typically corresponds to one shipping box, the total number of boxes entering the CycleBox system during the pilot is expected to fall within the same range.

Relevant Stakeholders

Logistics companies and e-commerce platforms (Such as Weee!, Amazon, Walmart) represent the most important industry partners for CycleBox. These organizations manage large-scale parcel delivery operations and continuously purchase corrugated packaging for shipping. A cardboard box return and reuse system can help logistics companies reduce packaging procurement costs while also supporting their sustainability and ESG goals. If logistics operators participate in pilot collaborations, the return flow of boxes can be integrated more efficiently with existing delivery networks.

Urban consumers are another critical participant in the system.

Most e-commerce shipping boxes are generated at the household level, which means consumer willingness to return boxes directly affects the feasibility of a reuse system. CycleBox introduces a combination of reward-based incentives and emotionally engaging design elements to encourage participation and motivate users to return boxes so they can re-enter the logistics cycle.

Community property managers and residential management organizations also play an important role during the pilot stage. Community spaces can serve as collection points for returned boxes, reducing operational costs associated with the reverse logistics process. Support from residential communities can help establish stable return channels and increase participation among residents.

Local governments and environmental organizations represent an additional group of external stakeholders. As cities face growing pressure in solid waste management, many public institutions are encouraging more efficient resource circulation systems. The whole-box reuse model proposed by CycleBox aligns with policy goals aimed at reducing packaging waste and improving material utilization efficiency.

Validation of Problem

CycleBox is currently in the early validation stage and has not yet conducted formal structured interviews with potential customers. However, the problem identification is based on clear observed behavior in existing e-commerce usage scenarios.

In current delivery practices, users typically receive packages, unpack them, and discard the boxes immediately. This suggests that while packaging waste is widespread, it is not actively addressed by users due to a lack of convenient return options and incentives.

To move beyond observation and begin direct validation, CycleBox has initiated outreach to potential partners. The team has identified Weee! as a suitable e-commerce platform for pilot collaboration and has begun contacting relevant personnel. At this stage, discussions are ongoing and responses are pending.

Solution

Proposed Solution Explained

On the user side, CycleBox increases participation in box returns by combining material incentives with emotional incentives. After receiving a package, users can return intact shipping boxes through simple channels, such as going to a community collection points or leaving it at the door waiting for a parcel pickup by delivery staff. In return, users receive platform-based rewards such as points, discounts, or other benefits provided through partnerships with e-commerce or logistics platforms.

These direct incentives help compensate for the time and effort required to return boxes, making participation more attractive to consumers.

Beyond financial rewards, CycleBox also incorporates emotionally engaging design and interactive feedback mechanisms to improve the user experience. Shipping boxes can feature distinctive visual designs, while digital platforms can provide positive feedback when users participate in the return process. These elements help make the experience more engaging and encourage users to develop consistent return habits over time.

On the logistics side, CycleBox does not attempt to build an entirely new transportation system. Instead, it integrates with existing delivery networks to create a practical return pathway for reusable boxes. During the early pilot stage, returned boxes can be collected through community drop-off locations or picked up by delivery personnel. Logistics partners can then conduct simple quality checks and redistribute suitable boxes back into the shipping process, reducing the need to purchase new packaging materials.

From an implementation perspective, CycleBox follows a phased rollout strategy. The first phase focuses on low-cost pilot programs in selected urban residential communities. These pilots aim to validate three key operational metrics: user participation in box returns, the percentage of boxes that meet reuse standards, and the average number of reuse cycles per box. Once these behavioral and operational patterns are validated, the second phase can explore structural improvements such as packaging standardization, box durability enhancements, and more systematic return infrastructure.

Value Proposition

CycleBox creates value by establishing a practical return pathway for shipping boxes and gradually shifting e-commerce packaging from a single-use model toward a circular reuse system. Through this approach, the solution aligns incentives across consumers, logistics operators, and urban communities while improving the overall efficiency of packaging circulation.

For logistics companies and e-commerce platforms, CycleBox offers a potential way to reduce packaging procurement costs. In current parcel delivery systems, shipping boxes represent a recurring operational expense. By increasing the reuse rate of intact boxes, logistics operators can reduce their dependence on continuously purchasing new packaging materials without significantly changing their existing delivery infrastructure. At the same time, this approach supports corporate sustainability initiatives and strengthens companies' ESG commitments.

For consumers, CycleBox transforms box returns from an inconvenient task into a rewarding and engaging activity. Through material incentives such as platform points, discounts, or other rewards, users receive direct benefits when returning reusable boxes.

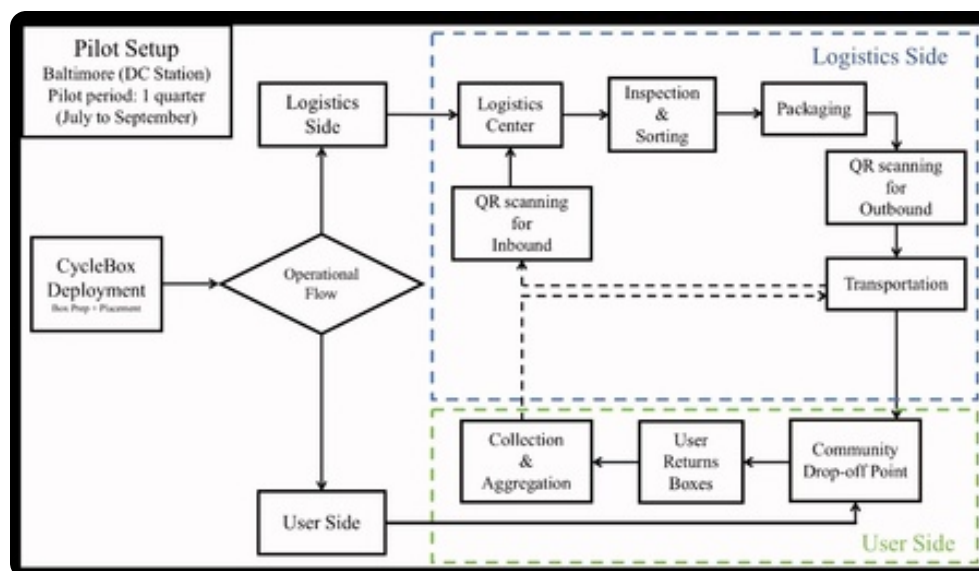
In addition, emotionally engaging visual design and interactive feedback mechanisms make the experience more enjoyable, encouraging users to participate more consistently over time.

Compared with existing solutions, CycleBox focuses on whole-box circulation rather than material recycling. Most current recycling systems treat cardboard boxes as raw material that is compressed and repulped, which eliminates the possibility of repeated use. CycleBox instead connects households, community collection points, and logistics networks to create a clearer pathway for boxes to return to the shipping system. This approach preserves the functional value of the box and improves overall resource efficiency.

Beyond the direct participants in the system, CycleBox also creates broader value for residential communities and local governments. By increasing the reuse rate of shipping boxes, the model can reduce the volume of packaging waste entering municipal recycling and waste management systems. This contributes to more efficient resource use and aligns with many cities' long-term sustainability and circular economy goals.

Solution Life Cycle

The CycleBox solution follows a closed-loop system that integrates both the logistics side and the user side within the existing delivery network.



On the logistics side, boxes are prepared and used during the normal fulfillment process. Each box is labeled with a barcode and tracked through scanning during outbound and inbound operations.

After packaging and dispatch, boxes are delivered to users through existing transportation and courier services.

On the user side, customers receive the package and unpack the items. Instead of discarding the box, they return it to designated collection points within the community. These collection points serve as temporary storage locations for returned boxes and are managed in coordination with community or property management.

Returned boxes are then transported back into the logistics system. They are scanned into inventory, inspected for basic quality, and prepared for reuse in future deliveries. Throughout this process, user participation and box movement are recorded through the digital system, enabling tracking and system optimization.

By embedding this process into the existing logistics infrastructure, CycleBox creates a continuous circulation loop without requiring a new transportation network. This allows packaging to be reused multiple times while maintaining operational efficiency.

Full Life Cycle Use Case

A typical use case of CycleBox can be illustrated through a student living near Johns Hopkins University ordering groceries from an e-commerce platform such as Weee!.

The process begins when the user places an order online. The items are delivered in a cardboard box through the existing logistics network. After receiving the package, the user unpacks the items and keeps the box in usable condition.

Instead of discarding the box, the user returns it through a nearby community collection point or hands it back during a future delivery. The return process is simple and integrated into the user's daily routine.

Once collected, the box re-enters the logistics system. It is inspected for basic quality and then redistributed for future deliveries. At the same time, the user receives points or small rewards through the platform, reinforcing participation.

Through this process, a single box can be used multiple times, reducing waste and improving the overall efficiency of packaging usage.

Why Our Solution is Superior

CycleBox's core advantage lies in addressing a key bottleneck in e-commerce packaging circulation:

user participation. In the current system, cardboard boxes are generally treated as single-use packaging materials. Even when they enter recycling channels, most are compressed and repulped rather than reused as intact shipping boxes. CycleBox introduces a return pathway that allows boxes that remain in good condition to re-enter the logistics system, improving the efficiency of packaging resource utilization.

Another advantage of CycleBox is the practical feasibility of its implementation. During the early stage, the project relies on existing e-commerce delivery networks and community collection points to conduct pilot programs. This approach avoids the need to build a completely new recycling infrastructure at the beginning. It also allows the system to test key operational indicators in real-life environments, including user return participation rates, the proportion of boxes suitable for reuse, and the average number of reuse cycles per box.

In addition, CycleBox incorporates a digital system to record user participation and support incentive mechanisms such as points rewards and interactive feedback. This system can encourage users to participate in box returns while also generating operational data that can be used to improve the efficiency of the circulation model over time.

As the return network gradually develops, CycleBox aims to establish stable circulation pathways for shipping boxes within specific communities. Once a functioning system of user participation and logistics coordination is established, new entrants would need to replicate the user engagement mechanism, the return pathway, and the supporting data system simultaneously. In practice, this creates a certain level of entry barrier for potential competitors.

Validation of Solution

CycleBox is currently in the early stage of solution validation and has not yet collected formal customer feedback or market traction data. However, the solution design is based on addressing clear gaps in existing user behavior and logistics systems.

The proposed solution focuses on reducing friction in box return by embedding the process into existing delivery infrastructure and providing simple collection points within communities. Compared to traditional recycling methods, CycleBox improves convenience and introduces incentives, which are key factors influencing user participation.

To validate the effectiveness of the solution, CycleBox has designed a three-month pilot in the Johns Hopkins University surrounding area. The pilot will test whether users are willing to participate in box return programs when convenience and incentives are provided.

Marketing Plan

Market Entry Strategy and Sales Plan

CycleBox will enter the market through a pilot program in the residential area surrounding Johns Hopkins University in Baltimore. CycleBox targets mid-scale e-commerce platforms such as Weee!, which operate frequent delivery services in dense urban areas. The pilot will run for one quarter and is expected to involve approximately 200 to 300 households. Based on typical e-commerce usage behavior, this pilot is estimated to generate between 5,000 and 9,000 boxes over the three-month period. This scale is sufficient to observe user participation patterns and operational performance in a real-world setting. Participation is introduced directly within the platform environment, allowing users who already receive packages to easily join the box return program. This approach reduces friction and improves adoption efficiency.

The key customers for CycleBox are e-commerce platforms and logistics operators. These organizations are responsible for packaging procurement and delivery operations, and they are the primary decision-makers for adopting the system. CycleBox approaches these partners by proposing a structured pilot, with clear performance metrics such as reuse rate, participation rate, and operational cost impact.

CycleBox provides an integrated service that connects user behavior, logistics operations, and data tracking. The system includes a digital platform for tracking returned boxes and managing incentives, along with a physical return process supported by community collection points and courier-assisted pickup during regular delivery routes. This design allows the system to operate within existing logistics networks.

Pricing is based on the value created through box reuse. The service is charged to partner platforms on a per reused box basis, with pricing set below the cost of purchasing new boxes. This ensures that partners can achieve direct economic benefits while participating in the system. During the pilot phase, pricing remains flexible and will be refined based on observed data.

The service is deployed within local residential communities. Return activities take place either at designated collection points or through direct pickup by delivery personnel. This setup keeps the process simple for users and aligns with existing delivery flows.

User engagement is driven through platform-based communication and incentives. Participation is encouraged through in-app notifications, reward points, and clear return instructions. The combination of convenience and tangible incentives helps build consistent user behavior over time.

This market entry approach focuses on validating the model in a controlled environment.

By concentrating on a defined geographic area and leveraging existing delivery infrastructure, CycleBox is able to test key assumptions and generate data to support future expansion.

Business Model

CycleBox operates as a service layer that connects end users, e-commerce platforms, and logistics systems to enable the reuse of shipping boxes. The model is built around integrating box return behavior into existing delivery workflows and converting that behavior into measurable operational value.

In this system, users receive packages through partner platforms and generate used cardboard boxes at home. CycleBox provides a simple return mechanism through community collection points or courier-assisted pickup during regular delivery routes. Returned boxes are collected, checked for usability, and reintroduced into the logistics system for future deliveries. A digital platform supports this process by tracking box movement, recording user participation, and managing incentive distribution.

Revenue is generated from e-commerce and logistics partners based on the number of boxes successfully reused. Each reused box represents avoided packaging procurement, creating direct cost savings for the partner. CycleBox captures a portion of this value through a per-box service fee. The pricing structure aligns with partner incentives by ensuring that the cost of using the service remains below the cost of purchasing new packaging.

In addition to per-box pricing, the model can be extended to include service-based agreements with larger partners. These agreements may cover system access, operational coordination, and data reporting. Data collected through the platform, including participation rates and reuse performance, also supports continuous optimization and provides partners with measurable sustainability metrics.

This model is designed to scale gradually. The initial pilot focuses on a defined community and leverages existing logistics infrastructure, allowing CycleBox to validate key assumptions while maintaining a low upfront investment. As participation increases and reuse networks become more stable, the system can expand to additional regions and partners.

Pricing

This is directly related to the cost savings generated by the reuse of cardboard boxes. The primary paying customers are e-commerce platforms and logistics operators, who are responsible for packaging procurement and delivery operations.

In a typical e-commerce environment, the cost of a brand-new cardboard box ranges from

approximately \$0.30 to \$0.80, depending on its size and specifications. By enabling box reuse, CycleBox helps its partners reduce the need to purchase new packaging. Therefore, its pricing is based on each reused box, meaning partners only pay when measurable value is created.

According to the pilot design in the residential area of Johns Hopkins University, the system is expected to process approximately 5,000 to 9,000 boxes within three months. Operational costs—including collection, processing, labeling, and system maintenance—are expected to be distributed across this volume, resulting in a moderate cost per box. Taking these factors into account, CycleBox plans to charge approximately \$0.50 to \$1.00 per reused box during the initial deployment phase.

In addition to per-box pricing, CycleBox may introduce service-based agreements for larger partners in the future. These agreements could include system integration, data reporting, and operational coordination. However, during the pilot phase, the focus remains on a simple and transparent usage-based pricing model to facilitate adoption and data collection.

At this stage, as the project is still in the pre-pilot phase, pricing has not yet been finalized through direct negotiations with customers. The purpose of the pilot is to test key assumptions related to participation rates, reuse rates, and operational costs. The data collected during this period will be used to refine pricing and ensure alignment with partner expectations and market conditions.

Validation of Pricing

The team has initiated outreach to potential partners, including contacts related to Weee!, and is currently awaiting further discussions and feedback. The proposed pricing is based on publicly available industry information and an initial estimation of operational costs, including box procurement, collection and handling processes, and basic system support. These assumptions provide a starting point for pricing design but have not yet been tested in a real operating environment.

During the pilot phase, pricing will be validated through actual implementation. The team will observe partner acceptance of the pricing structure, as well as their feedback on cost, operational feasibility, and perceived value. Key performance indicators such as participation rate, return rate, and reuse efficiency will also be used to assess whether the pricing model is sustainable.

Validation of Sales Plan

The team has identified e-commerce platforms and logistics operators, including Weee!, as initial target customers and has started outreach to explore potential collaboration. Validation will come from partner engagement, their willingness to participate in the pilot, and feedback on operational feasibility. These results will inform how the sales approach is refined and scaled in future stages.

Follow-on Markets

Following the pilot in the residential area surrounding Johns Hopkins University, CycleBox plans to expand gradually into larger urban markets. The initial expansion will focus on areas with similar characteristics, such as other university neighborhoods and high-density apartment communities. These locations typically have concentrated user bases and high e-commerce usage, making them suitable for replicating the participation patterns and operational processes validated during the pilot.

In terms of platform partnerships, CycleBox aims to extend beyond a single pilot partner to a broader set of e-commerce platforms with frequent delivery patterns. Platforms such as Weee! represent strong early-stage candidates due to their high delivery frequency and localized customer base. As the system matures, CycleBox will explore opportunities with larger platforms such as Amazon and Walmart, which operate at a much larger scale and generate substantial packaging demand. In addition, regional delivery platforms and daily-use e-commerce services may also serve as potential partners, further expanding the applicability of the system.

As geographic coverage and platform partnerships grow, CycleBox is expected to build a more stable and efficient box return network. Data collected during the pilot phase, including participation rates, reuse rates, and operational costs, will support the optimization of logistics processes and help improve system performance at scale. These data-driven insights will also strengthen CycleBox's ability to engage with larger partners and demonstrate measurable value.

From a timeline perspective, the initial pilot is designed to be completed within one quarter. If the pilot demonstrates strong performance, expansion into additional communities can begin within the following 6 to 12 months, with a gradual transition from community-level deployment to broader city-level coverage. In the longer term, the model can be replicated across multiple cities with similar urban and e-commerce characteristics.

Validation of Follow-on Markets

The team has identified e-commerce platforms and logistics operators, including Weee!, as initial target customers and has started outreach to explore potential collaboration. Validation will come from partner engagement, their willingness to participate in the pilot, and feedback on operational feasibility. These results will inform how the sales approach is refined and scaled in future stages.

Operational Plan

Critical Success Factors and Challenges

CycleBox will be implemented through a phased pilot approach rather than immediate large-scale deployment. The initial stage focuses on establishing a practical cardboard return pathway within a limited number of residential communities. In this stage, users who receive e-commerce packages can return intact boxes either through designated community collection points or through courier pickup during delivery routes.

A digital tracking system will record user participation and basic box information. When a user returns a box that remains structurally intact, the system records the action and provides a reward through platform points or similar incentives. These data records allow the team to observe real user behavior and evaluate whether incentive mechanisms effectively increase participation.

The recovered boxes will then re-enter the logistics process through simple screening procedures. Logistics partners can inspect returned boxes and determine whether they meet basic reuse standards. Boxes that pass this screening can be redistributed for future deliveries, completing a reuse cycle within the existing distribution network.

This pilot-driven implementation allows the project to test the operational feasibility of the system before considering larger-scale expansion.

Key Steps

The first key step is identifying a suitable pilot community where cardboard return points can be established and where a sufficient volume of e-commerce deliveries already occurs. A small geographic scope allows the team to observe user behavior and logistics interactions under realistic conditions.

The second step is launching the user participation mechanism. This includes introducing the reward system, enabling the digital recording of returned boxes, and communicating clearly to residents how they can participate in the recycling process.

The third step is integrating the return flow into the logistics system. Returned boxes must be collected, screened for structural integrity, and redistributed through existing delivery operations. Establishing this return flow is essential for closing the packaging loop.

The final step is evaluating pilot results. Key metrics include user participation rate, the proportion of boxes that meet reuse standards, and the average number of reuse cycles achieved per box. These indicators determine whether the system is viable for broader expansion.

Critical Success Factors

User participation is the most critical success factor for CycleBox. Since most e-commerce boxes are generated at the household level, the effectiveness of the entire system depends on whether users are willing to return boxes instead of discarding them. The incentive mechanism and ease of participation must therefore successfully encourage consistent user engagement.

Another key factor is the ability to integrate cardboard return flows into existing logistics operations. Because CycleBox relies on current delivery networks rather than building a new transport system, coordination with logistics partners is essential for enabling efficient box collection and redistribution.

A third success factor is the proportion of boxes that remain structurally suitable for reuse. Boxes that are heavily damaged or contaminated cannot re-enter the delivery cycle. Therefore, effective screening processes and appropriate reuse standards are necessary to maintain operational efficiency.

Challenges, Risks, and Mitigation

One potential challenge is low user participation during the early stage of the pilot. If too few users return boxes, the system may not generate enough reusable packaging to support a stable return flow. To mitigate this risk, CycleBox introduces direct incentives such as reward points and platform benefits to compensate users for the time and effort required to return boxes.

Another challenge involves the physical condition of returned boxes. Some boxes may become damaged during use or contaminated during disposal. Establishing clear reuse criteria and simple inspection procedures can help logistics partners identify boxes that remain suitable for reuse.

Operational coordination may also present challenges, particularly during the early stages when return flows are still being established. Maintaining a limited pilot scope and working within a single community helps reduce logistical complexity and allows the team to adjust operational processes gradually.

Timeline

The first stage focuses on preparation and pilot design, including selecting a suitable community and establishing cardboard return points. During this stage, the participation mechanism and digital tracking system are also prepared.

The second stage involves launching the pilot and observing real user participation. Data on return behavior, box condition, and logistics handling will be collected and analyzed.

The third stage evaluates the pilot results.

If participation rates and reuse indicators show promising outcomes, the system can begin exploring expansion to additional communities and further optimization of the packaging return network.

Key Assumptions

From a market perspective, the first assumption is that a meaningful proportion of users are willing to return intact cardboard boxes if the process is convenient and accompanied by direct incentives. Since most e-commerce boxes are generated at the household level, user participation is essential for creating a stable return flow. CycleBox assumes that incentive mechanisms such as reward points, discounts, or other platform benefits can encourage users to return boxes instead of discarding them.

A second market assumption is that logistics partners can integrate returned boxes into their existing delivery operations without significant disruption. Because CycleBox relies on current distribution networks rather than building a new transportation system, the project assumes that simple collection and screening processes can be incorporated into existing logistics workflows.

From an operational perspective, CycleBox assumes that a sufficient proportion of used cardboard boxes remain structurally suitable for reuse. Not all boxes will meet reuse standards due to damage or contamination, but the system assumes that enough boxes will remain usable to support repeated delivery cycles.

These assumptions will be validated through a pilot program conducted in a limited number of residential communities. During the pilot stage, CycleBox will measure three key indicators: the user participation rate in cardboard return, the proportion of returned boxes that meet reuse standards, and the average number of reuse cycles achieved per box. The results of these measurements will determine whether the system can operate effectively and whether broader implementation should be considered.

Barriers to Entry

First, CycleBox relies on the gradual formation of a stable return pathway between households, community collection points, and logistics networks. Establishing such a system requires coordination among multiple participants, including residents, community management, and delivery partners. Once these operational relationships and routines are in place, it becomes more difficult for new entrants to quickly build an equivalent return network in the same communities.

Second, the system depends on the development of consistent user participation. Encouraging households to return intact cardboard boxes requires behavioral change supported by incentives and repeated interaction with the platform. Over time, if users become accustomed to returning boxes through a specific system, it may be harder for alternative platforms to attract the same participation.

Third, CycleBox will accumulate operational data during the pilot and early deployment stages. Information such as participation rates, box reuse ratios, and average reuse cycles will help refine the operational model. This data can support improvements in logistics coordination and system design, potentially creating a learning advantage for the platform as it expands.

Finally, the early establishment of pilot programs may provide a first-mover advantage in specific communities. By initiating the cardboard return pathway and testing operational processes earlier, CycleBox can gain practical experience in implementing circular packaging systems in real urban environments.

Regulatory Requirements (if applicable)

Reimbursement Strategy (if applicable)

Intellectual Property (if applicable)

Costs and Financing

Operational Costs

CycleBox's operational costs during the pilot phase consist of platform development and maintenance, labor, operational execution, and box-level processing costs. These estimates are based on a three-month pilot conducted in the residential area surrounding Johns Hopkins University, with an expected scale of approximately 5,000 to 9,000 boxes.

First, in terms of digital infrastructure, CycleBox requires a basic online platform to track user participation, distribute incentives, and manage operational data. During the pilot phase, a simplified system is sufficient to support core functionalities. The initial development cost is estimated at approximately \$3,000 to \$8,000, with additional maintenance and hosting costs of around \$200 to \$600 over the three-month period.

Second, labor costs represent a key component of operations. These include project coordination, data tracking, and daily operational management. The pilot is designed to operate with a small, part-time team to control expenses. Based on typical hourly rates for student or entry-level operational roles, the total labor cost is estimated at approximately \$5,000 to \$9,000 for the three-month period.

In terms of operational execution, the main costs arise from organizing community collection points and coordinating box return and redistribution through existing delivery networks.

Because the system leverages current logistics infrastructure, no additional transportation system is required. As a result, incremental operational costs are relatively low and are estimated at approximately \$500 to \$1,500 for the pilot period.

In addition, each box involves a certain level of processing cost, including barcode stickers, basic protective coatings, and simple visual design elements. These components support box identification, improve durability, and enhance user experience. On a per-box basis, these costs are estimated at approximately \$0.10 to \$0.30 per box.

Finally, customer acquisition costs remain relatively low during the pilot phase. CycleBox relies on platform-based communication and community-level engagement rather than paid advertising. Most marketing-related expenses are associated with user incentives such as points or discounts, with total costs estimated at approximately \$1,000 to \$3,000 depending on participation levels.

The above cost ranges are derived from a combination of industry benchmarks, pilot-scale assumptions, and conservative estimation. Platform costs are based on typical pricing for small-scale web application development and cloud services. Labor costs are estimated using common hourly rates for part-time operational roles. Operational execution costs reflect the use of existing logistics networks, which significantly reduces infrastructure requirements. Box-level processing costs are based on standard bulk pricing for labeling and low-cost material treatments. Incentive costs are estimated based on common reward ranges used in behavior-driven engagement systems.

At the current pilot stage, CycleBox's unit economics are not yet optimized for profitability. Due to the limited scale of operations and the presence of fixed costs such as platform development and initial setup, the average cost per box remains relatively high when distributed across a smaller number of units.

However, the economic model is designed to improve with scale. As the number of participating users increases and the volume of reusable boxes grows, fixed costs can be spread across a larger base, reducing the average cost per box. In addition, increasing the number of reuse cycles per box further improves cost efficiency by extending the value generated from each unit.

Other Costs

For product testing, costs are associated with validating the collection process, user interaction, and operational setup in real-world conditions. This includes small-scale materials, setup adjustments, and on-site coordination. Based on typical pilot testing budgets for small projects, this cost is estimated at approximately \$500–\$1,200.

For market validation, costs primarily include user surveys, feedback collection, and participation

incentives. Assuming a small group of active users receiving incentives in the range of \$1–\$3 per participant, and accounting for partial participation, the total cost is estimated at approximately \$400–\$1,200. This estimate reflects common practice in early-stage user testing and engagement validation.

Administrative costs include basic software subscriptions, data management tools, and team collaboration platforms. These services are typically low-cost monthly subscriptions, and the total cost over a three-month period is estimated at approximately \$300–\$800, based on standard pricing for commonly used tools.

In addition, a contingency budget is included to account for unexpected expenses, such as minor compliance issues or operational adjustments. This buffer is estimated at approximately \$300–\$800, following standard budgeting practices for early-stage pilot projects.

Funding needed and use of funds

To execute a three-month pilot in the residential area surrounding Johns Hopkins University, CycleBox estimates that it will require approximately \$15,000 to \$22,000 in initial funding. This estimate reflects a realistic budget for a small-scale pilot in the U.S. context, balancing cost control with the need to generate reliable operational data.

The use of funds is structured across several key categories.

A portion of the budget will be allocated to platform development and maintenance, supporting the creation of a basic system for tracking user participation and managing incentives. Based on typical MVP development costs, this component is expected to account for approximately \$4,000 to \$6,000.

Labor represents the largest share of the budget, covering project coordination, operational management, and data tracking. With a small part-time team operating over three months, labor costs are estimated at approximately \$6,000 to \$10,000.

In addition, funds will be allocated to user incentives, such as points or discounts, which are critical for encouraging participation. Given expected participation rates and incentive levels, this component is estimated at approximately \$1,500 to \$4,000.

Operational execution and other supporting costs, including coordination, testing, and administrative expenses, are expected to total approximately \$2,000 to \$5,000.

Source of Funds

First, part of the initial funding will come from the founding team. This will be used to support early-stage setup, including basic system development and initial operational costs. This approach also shows the team's commitment to the project and helps build credibility when seeking external funding later.

Second, CycleBox will rely on support from university entrepreneurship programs and innovation competitions, such as incubators or startup challenges. These programs can provide initial funding, as well as mentorship and access to relevant networks, which are well aligned with the needs of a pilot-stage project.

In addition, CycleBox will explore small-scale funding opportunities related to sustainability, such as environmental grants or circular economy programs. Since the project focuses on reducing packaging waste and improving resource efficiency, it fits well with the goals of these funding sources.

Revenue Timeline

During the initial three-month pilot phase, CycleBox does not expect to generate revenue. The primary goal at this stage is to validate key assumptions, including user participation, box return rates, and operational feasibility, rather than to achieve immediate monetization.

If the pilot demonstrates strong results, CycleBox expects to begin generating initial revenue within 3 to 9 months after the pilot. This would come from early-stage partnerships with e-commerce platforms or logistics providers, where the service can be introduced on a small scale.

CycleBox expects to reach cash flow positivity in approximately 9 to 18 months, depending on the pace of expansion and the efficiency of operations. At this stage, a larger user base and more stable operational processes would allow the system to scale and reduce per-unit costs.

Team

Founding Team and Team Experience

The CycleBox team consists of four members: Haiyang Wu, Lingsheng, Wandong, and Siying, each contributing to different aspects of the project.

Haiyang Wu focuses on data analysis and overall system design. His role includes structuring the pilot, defining key metrics, and supporting decision-making based on data.

Lingsheng is responsible for external communication and outreach, including connecting with potential partners and supporting collaboration opportunities.

Wandong focuses on market research, including understanding industry practices, benchmarking existing solutions, and identifying potential application scenarios for CycleBox.

Siyang is responsible for coordination and organization, ensuring that project tasks are structured, documented, and executed efficiently across the team.

Advisors

Round 1: Alexei Yukna